ל3 — Identifying 3: Character Orthogonality on the Conductor Lattice

Leonardo Murillo Montero

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Abstract

Situation. ל1 gave the chart and the rotation group at the uniform. ל2 gave the cumulant tower — position, metric, cubic vertex, closure defect — generated by one functional $\Phi = \log Z$ on H_0. The cubic vertex T_{ijk} is a symmetric 3-tensor on a vector space of dimension $N-1$. As written, it is dense: every combination of indices contributes generically. Yet H_0 has more structure than its dimension. When the index set $\{1, \dots, N\}$ carries the additive structure of $\mathbb{Z}/N$, there is a canonical basis adapted to that structure, and the cumulant tensors decompose along it.

Task. Make the arithmetic structure of H_0 visible. Choose a basis on H_0 that diagonalizes the natural cyclic action and read ל2's tensors in it.

Action. Identify the outcome index set with $\mathbb{Z}/N$. Take the Pontryagin dual $\widehat{\mathbb{Z}/N}$ — the characters $\chi_k(x) = e^{2\pi i k x/N}$. Use character orthogonality to decompose H_0 into **conductor packets** H_d, one for each non-trivial divisor $d \mid N$.

Result. Two facts about the cumulant tower at the uniform, read in the character basis:

- Σ at the uniform is **scalar** in the character basis, hence trivially block-diagonal across conductor packets. Order 2 carries no cross-packet content.
- T at the uniform satisfies the **selection rule** $a + b + c \equiv 0 \pmod N$: the cubic vertex of three non-trivial characters is non-zero exactly when their indices sum to zero modulo N. Order 3 couples packets across divisors of N.

This is the third — and decisive — appearance of *Identifying 3*: order 2 is arithmetically diagonal; order 3 introduces the first finite arithmetic constraint, the selection rule, that the ring structure of $\mathbb{Z}/N$ imposes on the cumulant tower. The single line $k + \ell + m \equiv 0 \pmod N$ is the load-bearing arithmetic of the trilogy.

This is ל3 of the trilogy *Identifying 3*. ל1 supplied the chart; ל2 supplied the tower; ל3 supplies the arithmetic substrate on which both can be read.

Keywords: information geometry, categorical distribution, Pontryagin duality, character orthogonality, discrete Fourier transform, conductor, Dirichlet characters, Amari–Chentsov tensor, cumulant tower, selection rule, cyclic group, Euler totient.

Label the N outcomes of a categorical distribution by something cyclic (days of the week, hours of the day, decimal digits, residues modulo a prime) and the natural basis on the simplex is no longer the labels themselves but their discrete Fourier modes, indexed by a frequency that wraps around at N. Read the cumulant tower in that basis. The Fisher information metric becomes the identity: arithmetic is silent at order 2. The cubic vertex of three modes is non-zero exactly when their three frequencies add to zero modulo N. That single line, $a+b+c \equiv 0 \pmod N$, is the same arithmetic that physicists call momentum conservation on a lattice, optical engineers call three-wave mixing, and number theorists know from the convolution algebra of cyclic-group characters. The third cumulant is where the ring structure of the labels first speaks; one finite, decidable equation closes the trilogy on its third and final '3'.

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Notation (extending לּ1, לּ2)

Symbol	Meaning
$\mathbb{Z}/N$	The cyclic ring of integers modulo N; the outcome index set with its additive structure (§1).
$\widehat{\mathbb{Z}/N}$	The Pontryagin dual: the group of characters of $\mathbb{Z}/N$ (§2).
χ_k	The character $\chi_k(x) = e^{2\pi i k x/N}$, indexed by $k \in \mathbb{Z}/N$.
$\mathrm{ord}(\chi_k)$	The order of χ_k in $\widehat{\mathbb{Z}/N}$: smallest $m \geq 1$ with $\chi_k^m = 1$. Equals $N/\gcd(k, N)$.
$\mathrm{cond}(\chi_k)$	The **additive conductor** (a.k.a. minimal quotient modulus) of χ_k: equals $\mathrm{ord}(\chi_k)$ in our setting. Named by analogy with — not by definitional identity to — the Dirichlet-character conductor; see §3.
H_d	The conductor-d packet: $\mathrm{span}_{\mathbb{C}}\{\chi_k : \mathrm{cond}(\chi_k) = d\}$, for divisor $d \mid N$ (§3).
$\varphi(d)$	Euler totient: count of $k \in \{0, \ldots, d-1\}$ coprime to d; equals $\dim_{\mathbb{C}} H_d$.

Notation from לּ1 (Δ^{N-1}, H_0, u) and לּ2 (Φ, Σ_{ij}, T_{ijk}, $Q_{ijk\ell}$) remains in force.

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1. Ring structure on the outcome index

לּ1 and לּ2 treated the outcome index set $\{1, \ldots, N\}$ as an unstructured finite set. The simplex Δ^{N-1}

is invariant only under the symmetric group S_N (relabelings), and the cumulant tower respects nothing finer than S_N.

Many naturally arising categorical settings, however, equip $\{1, \dots, N\}$ with **more structure** — a cyclic ordering, a ring, or a group law on the labels. Examples:

- **Days of the week** ($N = 7$): the labels carry the additive structure of $\mathbb{Z}/7$.
- **Months of the year** ($N = 12$): the labels carry $\mathbb{Z}/12$.
- **Decimal digits** ($N = 10$): the labels carry $\mathbb{Z}/10$.
- **Hours of the day** ($N = 24$): the labels carry $\mathbb{Z}/24$.
- **Residues modulo a fixed prime** p: the labels carry $\mathbb{Z}/p$, with multiplicative as well as additive structure.

Whenever such structure is given (or chosen), the natural action on the simplex contains a cyclic subgroup $\mathbb{Z}/N \subset S_N$ acting by index rotation, and the relevant symmetry group of H_0 contracts from S_N to a quotient that respects the cyclic order.

ל3 makes the following stipulation: **the outcome index set is** $\mathbb{Z}/N$, the cyclic ring of integers modulo N. With this stipulation, H_0 inherits an action of $\mathbb{Z}/N$ (by cyclic shift) and a Pontryagin dual.

The stipulation is not automatic. It is the choice ל3 makes, and the entire arithmetic apparatus that follows depends on it. For unstructured index sets, ל2's tensors are the final word; for $\mathbb{Z}/N$-structured ones, ל3 supplies a decomposition ל2 cannot see.

2. Pontryagin duality and the character basis

The **Pontryagin dual** of $\mathbb{Z}/N$ is the group of characters $\chi : \mathbb{Z}/N \to \mathbb{C}^\times$ (homomorphisms to the multiplicative group of non-zero complex numbers). Every such character has the form

$$\chi_k(x) = e^{2\pi i k x/N}, \qquad k \in \{0, 1, \dots, N-1\},$$

and the map $k \mapsto \chi_k$ identifies $\widehat{\mathbb{Z}/N}$ with $\mathbb{Z}/N$ itself.

Character orthogonality. The fundamental identity is

$$\frac{1}{N} \sum_{x \in \mathbb{Z}/N} \chi_k(x)\, \overline{\chi_\ell(x)} = \delta_{k,\ell}, \tag{1}$$

equivalently: the N characters $\{\chi_k\}_{k=0}^{N-1}$ are orthonormal in the standard Hermitian inner product on $\mathbb{C}^N$ (scaled by $1/N$). They therefore form a **unitary basis** of $\mathbb{C}^N$.

The trivial character $\chi_0(x) = 1$ for all x spans the constant subspace — vectors with all entries equal, which lie outside H_0. The remaining $N-1$ characters $\chi_1, \dots, \chi_{N-1}$ span the complexification of H_0:

$$H_0 \otimes_{\mathbb{R}} \mathbb{C} = \operatorname{span}_{\mathbb{C}}\{\chi_1, \dots, \chi_{N-1}\}.$$

For real vectors, the conjugation $v \mapsto \overline{v}$ acts on character coefficients as $k \leftrightarrow -k \pmod N$, so real vectors in H_0 are characterized by $\hat{v}_{-k} = \overline{\hat{v}_k}$ where $\hat{v}_k = \frac{1}{N}\sum_x \overline{\chi_k(x)}\, v_x$ is the k-th character coefficient.

Real basis. When working over $\mathbb{R}$ is preferred, pair conjugate characters χ_k and χ_{N-k} into real combinations

$$c_k(x) = \cos\!\left(\tfrac{2\pi kx}{N}\right), \qquad s_k(x) = \sin\!\left(\tfrac{2\pi kx}{N}\right), \qquad 1 \le k < N/2,$$

each pair spanning a 2-dimensional real subspace of H_0. If N is even, the self-conjugate character $\chi_{N/2}(x) = (-1)^x$ is real-valued and contributes a 1-dimensional real subspace on its own. The trivial character χ_0 is the constant subspace and excluded from H_0. Dimension count: $2 \cdot (\lfloor (N-1)/2 \rfloor) + [N \text{ even}] = N - 1 = \dim_{\mathbb{R}} H_0$ ✓.

For the rest of §3 we work over $\mathbb{C}$, where the character basis is unitary and the algebra is cleanest; results restrict canonically to the real form via the conjugation rule above.

> **Aside.** The character basis is exactly the **discrete Fourier transform** of $\mathbb{Z}/N$. Readers from signal processing can read the entire chapter as "Fourier-diagonalize the cumulants of the categorical distribution"; readers from number theory can read it as an additive analogue of Dirichlet-character decomposition for cyclic Gibbs ensembles (additive characters of $\mathbb{Z}/N$, not multiplicative Dirichlet characters on $(\mathbb{Z}/N)^\times$ — see §3). The two perspectives are equivalent.

3. Conductor packets

The character χ_k has a well-defined **order** in the group $\widehat{\mathbb{Z}/N}$ — the smallest $m \ge 1$ such that $\chi_k^m = \chi_0$. Direct computation:

$$\chi_k^m(x) = \chi_{km}(x) = e^{2\pi i k m x/N} = 1 \text{ for all } x \iff N \mid km.$$

The smallest such m is $m = N/\gcd(k, N)$. Define this as the **conductor** of χ_k:

$$\operatorname{cond}(\chi_k) := \operatorname{ord}(\chi_k) = \frac{N}{\gcd(k, N)} \in \{d : d \mid N\}.$$

Equivalently, $\operatorname{cond}(\chi_k)$ is the smallest divisor d of N such that χ_k factors through the quotient map $\mathbb{Z}/N \twoheadrightarrow \mathbb{Z}/d$ — the smallest d on which χ_k is "natively defined." We call this the **additive conductor** of χ_k, or, more neutrally, its **minimal quotient modulus**.

On the terminology "conductor." The label is used here **by analogy** with the Dirichlet-character conductor of analytic number theory, not as the same definition. The Dirichlet setup is multiplicative: a Dirichlet character $\chi \pmod N$ is a homomorphism $(\mathbb{Z}/N)^\times \to \mathbb{C}^\times$ on the *units*, and it is *induced* from a smaller modulus f if a primitive multiplicative $\chi^* \pmod f$ exists with $\chi(x) = \chi^*(x)$ whenever $\gcd(x, N) = 1$; the smallest such f is the (multiplicative) conductor. The characters χ_k in this paper are *additive* characters of the abelian group $\mathbb{Z}/N$, defined on every element (units and non-units alike), and the "induction" we use is factorization through a group quotient $\mathbb{Z}/N \twoheadrightarrow \mathbb{Z}/d$. Multiplicative Dirichlet characters and additive characters of $\mathbb{Z}/N$ are not dual bases of the same space, and the two notions of "conductor" are not literally the same definition.

In the restricted setting of additive characters of a cyclic group both attach the same divisor of N — namely $N/\gcd(k,N)$, the order of χ_k — and that coincidence is what justifies recycling the word. Throughout the rest of §3 and the trilogy, $\mathrm{cond}(\chi_k)$ means the **additive conductor / minimal quotient modulus** defined above.

Definition (conductor packet). For each divisor $d \mid N$,

$$H_d = \mathrm{span}_{\mathbb{C}}\{\chi_k : \mathrm{cond}(\chi_k) = d\} \subset \mathbb{C}^N.$$

The number of characters of conductor exactly d is the number of $k \in \{0, \dots, N-1\}$ with $\gcd(k,N) = N/d$, which equals Euler's totient $\varphi(d)$. So $\dim_{\mathbb{C}} H_d = \varphi(d)$.

Decomposition. Summing over divisors:

$$\mathbb{C}^N = \bigoplus_{d|N} H_d, \qquad H_0 \otimes_{\mathbb{R}} \mathbb{C} = \bigoplus_{\substack{d|N \\ d>1}} H_d. \tag{2}$$

The trivial packet $H_1 = \mathrm{span}\{\chi_0\}$ is the constant subspace and is excluded from H_0. Throughout the rest of §3, every H_d referenced (for $d > 1$) sits inside the complexification of H_0; we write $H_d \subset H_0 \otimes \mathbb{C}$ when emphasis matters and H_d alone otherwise. The dimension count $\sum_{d|N} \varphi(d) = N$ is the classical Euler identity, and removing H_1 leaves $N - 1 = \dim H_0$.

Picture for $N = 6$. Divisors of 6: $\{1, 2, 3, 6\}$. The character lookup table:

k	$\gcd(k,6)$	$\mathrm{cond}(\chi_k) = 6/\gcd$	Packet
0	6	1	H_1 (trivial)
1	1	6	H_6
2	2	3	H_3
3	3	2	H_2
4	2	3	H_3
5	1	6	H_6

So H_0 for $N = 6$ decomposes as $H_2 \oplus H_3 \oplus H_6$, with $\dim H_2 = \varphi(2) = 1$, $\dim H_3 = \varphi(3) = 2$, $\dim H_6 = \varphi(6) = 2$. Total: $1 + 2 + 2 = 5 = 6 - 1$ ✓.

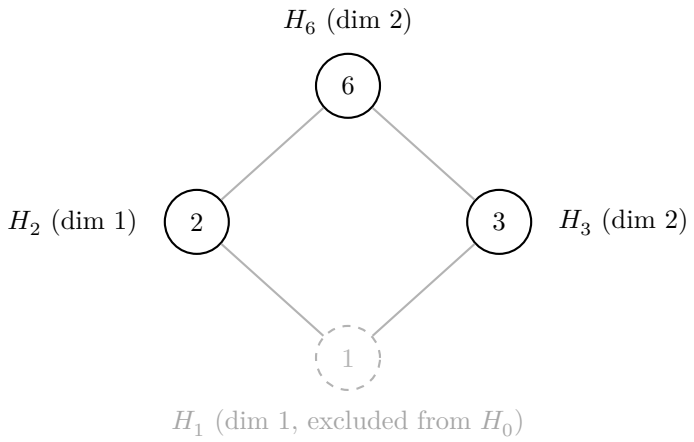


Figure 1: The divisor lattice and conductor packets of $N = 6$. Each node d carries the conductor packet $H_d \subset \mathbb{C}^N$ of characters χ_k with conductor exactly d, of complex dimension $\varphi(d)$. Cover relations are $1 < 2$, $1 < 3$, $2 < 6$, $3 < 6$. The trivial packet H_1 at the bottom (dashed) is the constant subspace and lies outside H_0; the decomposition $H_0 = H_2 \oplus H_3 \oplus H_6$ is the union of the three upper nodes. The divisor lattice is the combinatorial substrate of the packet decomposition.

4. The Fisher metric in the character basis

§2 gave $\Sigma_{ij}|_u = p_i\delta_{ij} - p_i p_j$. At the uniform $p_i = 1/N$:

$$\Sigma_{ij}|_u = \frac{1}{N}\delta_{ij} - \frac{1}{N^2}.$$

Σ is a symmetric bilinear form on the real space H_0. To read it in the character basis we evaluate it on character vectors. Two natural conventions are available:

- **Bilinear pairing.** $\Sigma(\chi_a, \chi_b) := \sum_{i,j} \Sigma_{ij}|_u \, \chi_a(i)\, \chi_b(j)$. Compatible with the orthogonality identity $\sum_x \chi_k(x)\chi_\ell(x) = N\delta_{k+\ell\equiv 0}$.
- **Hermitian pairing.** $\Sigma_h(\chi_a, \chi_b) := \sum_{i,j} \Sigma_{ij}|_u \, \chi_a(i)\, \overline{\chi_b(j)}$. Compatible with the orthonormality identity $\sum_x \chi_k(x)\,\overline{\chi_\ell(x)} = N\delta_{k,\ell}$.

The two are related by $\Sigma_h(\chi_a, \chi_b) = \Sigma(\chi_a, \chi_{-b})$, since $\overline{\chi_b} = \chi_{-b}$. Both lead to the same conclusion; we use Hermitian pairing throughout, which is the standard convention for complex orthonormal bases.

Compute:

$$\Sigma_h(\chi_a, \chi_b)\big|_u = \frac{1}{N}\sum_i \chi_a(i)\,\overline{\chi_b(i)} - \frac{1}{N^2}\sum_i \chi_a(i) \cdot \sum_j \overline{\chi_b(j)}.$$

Using $\sum_i \chi_a(i)\,\overline{\chi_b(i)} = N\delta_{a,b}$ and $\sum_i \chi_a(i) = N\delta_{a\equiv 0}$:

$$\Sigma_h(\chi_a, \chi_b)\big|_u = \delta_{a,b} - \delta_{a\equiv 0}\,\delta_{b\equiv 0}. \tag{3}$$

Restricting to H_0 means restricting to non-trivial characters ($a, b \not\equiv 0$), which forces the second term to vanish:

$$\Sigma_h(\chi_a, \chi_b)\big|_u = \delta_{a,b} \quad \text{for } a, b \in \{1, \dots, N-1\}. \tag{4}$$

Σ at the uniform is the identity in the character basis on H_0. It is therefore diagonal, in fact scalar — and trivially block-diagonal across conductor packets, since the diagonal partitions any way you like.

The order-2 content of the cumulant tower carries **no cross-packet structure**.

5. The Amari–Chentsov tensor in the character basis

↳2 equation (3) at $p_i = 1/N$:

$$T_{ijk}|_u = \frac{1}{N}\delta_{ij}\delta_{ik} - \frac{1}{N^2}(\delta_{ij} + \delta_{ik} + \delta_{jk}) + \frac{2}{N^3}.$$

T is a symmetric trilinear form on H_0 — a function $T : H_0 \times H_0 \times H_0 \to \mathbb{R}$ symmetric under any permutation of arguments. We evaluate it on three character vectors (χ_a, χ_b, χ_c) by contraction (using the bilinear pairing, since T is symmetric and the result is the most transparent in this convention; the Hermitian variant is obtained by conjugating one or three arguments):

$$T(\chi_a, \chi_b, \chi_c) := \sum_{i,j,k} T_{ijk}|_u \, \chi_a(i)\, \chi_b(j)\, \chi_c(k).$$

There are five contributions, one per term in the formula:

Term 1. $\frac{1}{N}\sum_{i,j,k} \delta_{ij}\delta_{ik}\, \chi_a(i)\chi_b(j)\chi_c(k) = \frac{1}{N}\sum_i \chi_a(i)\chi_b(i)\chi_c(i) = \frac{1}{N}\sum_i \chi_{a+b+c}(i) = \delta_{a+b+c\equiv 0}$.

Term 2. $-\frac{1}{N^2}\sum_{i,j,k} \delta_{ij}\, \chi_a(i)\chi_b(j)\chi_c(k) = -\frac{1}{N^2}\big[\sum_i \chi_{a+b}(i)\big]\big[\sum_k \chi_c(k)\big] = -\delta_{a+b\equiv 0}\,\delta_{c\equiv 0}$.

Terms 3, 4. Analogously: $-\delta_{a+c\equiv 0}\,\delta_{b\equiv 0}$ and $-\delta_{b+c\equiv 0}\,\delta_{a\equiv 0}$.

Term 5. $\frac{2}{N^3}\sum_{i,j,k} \chi_a(i)\chi_b(j)\chi_c(k) = 2\,\delta_{a\equiv 0}\,\delta_{b\equiv 0}\,\delta_{c\equiv 0}$.

Restriction to H_0. The trivial character χ_0 spans the constant subspace, which lies outside H_0. Restricting T to its natural domain — the cubic vertex on H_0 — therefore means restricting to non-trivial characters ($a, b, c \not\equiv 0 \pmod N$). Under this restriction every $\delta_{a\equiv 0}$, $\delta_{b\equiv 0}$, $\delta_{c\equiv 0}$ factor vanishes, killing Terms 2–5 in their entirety. Only Term 1 survives:

$$\boxed{T(\chi_a, \chi_b, \chi_c)\big|_u = \delta_{a+b+c\equiv 0 \pmod N} \quad \text{for } a, b, c \in \{1, \dots, N-1\}.} \tag{5}$$

This is the **selection rule**: at the uniform, the Amari–Chentsov tensor evaluated on three non-trivial characters is non-zero iff their indices sum to zero modulo N.

The rule is exact, finite, and decidable. Given any three character indices (a, b, c), the condition $a+b+c \equiv 0 \pmod N$ either holds or does not, and the tensor coefficient is either 1 or 0 accordingly.

Cross-packet coupling. Unlike Σ, the selection rule (5) couples characters from **different conductor packets**. Example for $N = 6$: take $a = 1$ (in H_6), $b = 2$ (in H_3), $c = 3$ (in H_2). Then

$a + b + c = 6 \equiv 0 \pmod 6$, so $T(\chi_1, \chi_2, \chi_3) = 1$ — a non-zero coupling between three different packets H_6, H_3, H_2. The cubic vertex sees the full divisor lattice of N.

6. The arithmetic content of the "leap to 3"

The trilogy's three appearances of *Identifying 3* are now in hand:

- **ל1 — coordinates.** The third cumulant is the first invariant that does not lift through the square-root map to a Riemannian sphere quantity; information geometry separates from sphere geometry at order 3.
- **ל2 — calculus.** $T_{ijk} = \partial^3 \Phi$ is the first cumulant that does not vanish for Gaussians; the cumulant tower's content beyond Riemannian geometry begins at order 3.
- **ל3 — arithmetic.** Σ at the uniform is scalar in the character basis (no arithmetic content); T at the uniform carries the selection rule $a+b+c \equiv 0 \pmod N$ (the first arithmetic content). Order 3 is where the ring structure of $\mathbb{Z}/N$ enters the cumulant tower.

In all three readings, the third object is where the content lives. The arithmetic reading is the sharpest: a single finite line — the modular equation $a + b + c \equiv 0 \pmod N$ — distinguishes the cubic vertex from the metric.

The selection rule is *necessary* for order-3 information geometry to be sensitive to the ring structure of $\mathbb{Z}/N$, and it is *sufficient*: every other cumulant tensor on the categorical family at the uniform, when restricted to non-trivial characters, factors through some condition of this kind on character indices. ל3's claim is the precise form of the order-3 case.

7. Worked examples

7.1. $N = 4$

Divisors: $\{1, 2, 4\}$. Packet structure on H_0:

Packet	Conductor	Generators	$\dim_{\mathbb{C}}$
H_2	2	χ_2	1
H_4	4	χ_1, χ_3	2

Selection rule: $T(\chi_a, \chi_b, \chi_c) \neq 0$ iff $a + b + c \equiv 0 \pmod 4$ and $a, b, c \in \{1, 2, 3\}$.

Triples satisfying the rule (up to ordering):

- $(1, 1, 2)$: $4 \equiv 0$ ✓ — couples H_4, H_4, H_2.
- $(2, 2, 0)$: 0-character excluded.
- $(1, 3, 0)$: 0-character excluded.
- $(2, 1, 1)$: same as $(1, 1, 2)$.
- $(3, 3, 2)$: $8 \equiv 0$ ✓ — couples H_4, H_4, H_2.

So the only non-zero cubic vertex on $N = 4$ at the uniform (up to symmetry) is $T(\chi_a, \chi_b, \chi_2)$ for $\{a, b\} = \{1, 1\}$ or $\{3, 3\}$. The cubic structure on $\mathbb{Z}/4$ is sparse — only one packet combination contributes.

7.2. $N = 6$

Divisors: $\{1, 2, 3, 6\}$. Packet structure on H_0 (recapped from §3):

Packet	Conductor	Generators	$\dim_{\mathbb{C}}$
H_2	2	χ_3	1
H_3	3	χ_2, χ_4	2
H_6	6	χ_1, χ_5	2

Definition (single-packet vs mixed-packet vertex). A triple (a, b, c) satisfying $a + b + c \equiv 0 \pmod N$ is called **single-packet** if $\mathrm{cond}(\chi_a) = \mathrm{cond}(\chi_b) = \mathrm{cond}(\chi_c)$, and **mixed-packet** otherwise. Single-packet vertices couple a packet to itself; mixed-packet vertices couple distinct packets.

For $N = 6$, triples (a, b, c) with $a + b + c \equiv 0 \pmod 6$ and $a, b, c \in \{1, 2, 3, 4, 5\}$, up to permutation:

Triple (sorted)	Sum	Packets	Type
$(2, 2, 2)$	6	H_3, H_3, H_3	single-packet (H_3)
$(4, 4, 4)$	12	H_3, H_3, H_3	single-packet (H_3)
$(1, 1, 4)$	6	H_6, H_6, H_3	mixed
$(2, 5, 5)$	12	H_3, H_6, H_6	mixed
$(1, 2, 3)$	6	H_6, H_3, H_2	mixed (three different)
$(3, 4, 5)$	12	H_2, H_3, H_6	mixed (three different)

So $N = 6$ supports both single-packet cubic vertices (only within H_3) and mixed-packet vertices coupling two or three distinct packets. This is the smallest squarefree N with more than one prime divisor — the minimal "mixed" ring on which the selection rule has non-trivial cross-packet structure.

The detailed accounting of which triples live in which packet combination — single-packet vs. mixed — is the basic combinatorial input to downstream work. ל3 establishes the rule (5); applications elsewhere read its consequences.

8. The toolkit, as a working list

Object	Without $\mathbb{Z}/N$ structure	With $\mathbb{Z}/N$ structure
Basis of H_0	Helmert frame (ל1 §7.3) or any orthonormal frame	Character basis $\{\chi_k\}_{k \neq 0}$
Tangent isometry group at u	$\mathrm{SO}(H_0)$ (ל1 §5; linear isometries of $g_{FR}\|_u$ on H_0, not the global isometry group of Δ^{N-1})	$\mathrm{SO}(H_0)$ restricted to commute with $\mathbb{Z}/N$ rotation
Σ at u	Scalar (ל2 §5.2)	Scalar in character basis — diagonal across packets
T at u	Dense symmetric 3-tensor (ל2 §5.2)	Selection rule: $a + b + c \equiv 0 \pmod N$

Object	Without $\mathbb{Z}/N$ structure	With $\mathbb{Z}/N$ structure
Packet decomposition	None	$H_0 = \bigoplus_{d\|N,\, d>1} H_d$
Dim of packet H_d	—	$\varphi(d)$ over $\mathbb{C}$

The order-2 row says: arithmetic does not enter at order 2.

The order-3 row says: arithmetic enters at order 3, in the form of one modular equation.

Three load-bearing identifications, three appearances of "3": - order 2 $\to$ order 3 in the cumulant tower (ל2); - $N = 2 \to N \geq 3$ in the simplex dimension (ל2 §4); - Σ-diagonal $\to$ T-modular-coupled in the conductor packet basis (ל3).

What the trilogy hands forward

The trilogy *Identifying 3* is complete. ל1 supplied the chart: the native simplicial coordinatization with H_0 as tangent space and $\mathrm{SO}(H_0)$ as the tangent (isotropy) isometry group at the uniform — the full linear isometry group of $g_{FR}|_u$ on $T_u\Delta^{N-1}$, not the global isometry group of (Δ^{N-1}, g_{FR}). ל2 supplied the tower: one functional $\Phi = \log Z$ whose successive cumulants generate position, metric, cubic vertex, and closure defects. ל3 supplied the substrate: the Pontryagin character basis on $\mathbb{Z}/N$ that diagonalizes the metric and constrains the cubic vertex by the single modular equation

$$a + b + c \equiv 0 \pmod{N}.$$

Three identifications, one workspace. Any downstream theory on categorical exponential families with $\mathbb{Z}/N$-structured outcomes now has:

- A coordinate-free vector space (H_0) to live on, with an intrinsic Euclidean algebra.
- A canonical successor tower of multilinear forms ($\Sigma, T, Q, \ldots$) on that space.
- A finite arithmetic decomposition (conductor packets H_d) with explicit selection rules on the multilinear forms.

The selection rule $a + b + c \equiv 0 \pmod{N}$ is the load-bearing arithmetic of the program. It is finite, decidable, and exact; it lives at exactly the cumulant order — three — where information geometry separates from Riemannian geometry; and it is the single line of modular arithmetic that the rest of the program reads as physics, as machine learning, and as number theory.

The trilogy ends here. The substrate is in place.

9. Acknowledgments

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